Analyse Data 15

Spreadsheets are used for automatic calculations. In the last lesson, we learnt how to use formulas for addition ('+', '/', SUM).

Apart from calculations, large sets of data can be arranged to be easier to understand. Conditional formatting, sort and filter are some features used in a spreadsheet. These help to sort data and find it quickly.

Conditional formatting

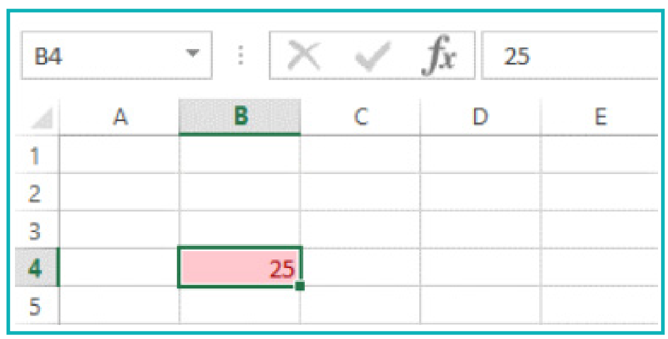
- Conditional formatting lets us apply specific formatting to cells if a certain condition is met.
- This feature checks whether the data meets the criteria specified. Then it automatically applies the formatting, such as cell colour.
- At times, we need to differentiate between sets of data on a spreadsheet. Conditional formatting can be used to highlight or colour cells which meet a particular criteria. This makes it easier for us to view and analyse different data.

1 **If–then logic:** If a certain condition is true, then a specific formatting will be applied to cells which meet the condition.

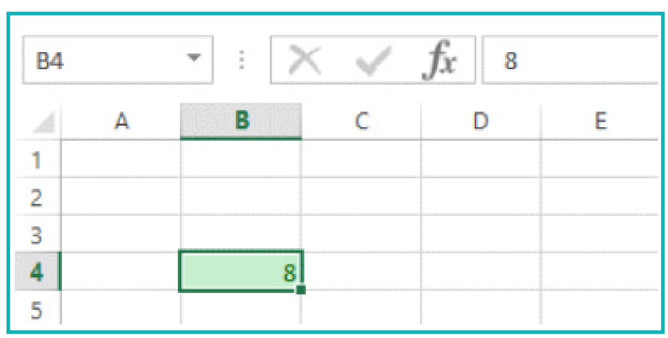
For example:

Here is a list of data that has numbers which range from 1 to 30.
Let's apply some conditions to the spreadsheet:

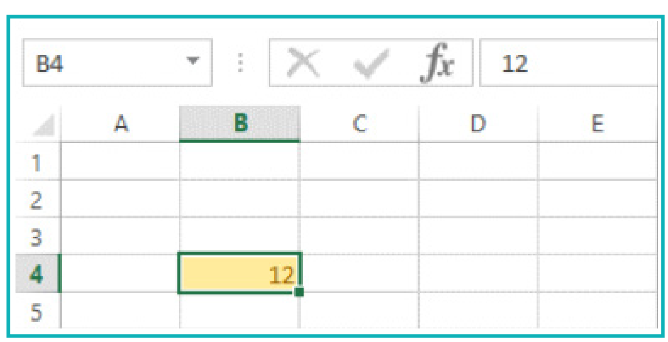
Greater than: If the value of cell B4 is **greater than 20,** then change the cell background to **Red.**



Less Than: If the value of cell B4 is **less than 10,** then change the cell background to **Green.**



Between: If the value of cell B4 is **greater than 10** and **less than 20,** then change the cell background to **Yellow.**



2 **Red–Yellow–Green colour scale:** When the **Red–Yellow–Green colour scale** is applied to a cell, Excel automatically calculates values and allots colours to cells:

- The cell that contains the maximum value is coloured red.
- The cell that contains the value in between is coloured yellow.
- The cell that contains the minimum value is coloured green.

Student Name	No. of Chocolates
Shivani	10
Kishor	6
Veena	9

Sort

- Sort is a process that arranges data into some kind of order. This makes it easier to understand, analyse, or visualise.
- Data can be arranged alphabetically (A to Z or Z to A) or numerically (smallest to largest or largest to smallest).
- At times, cell values are textual or numerical and need to be arranged in a sequence. This is usually done to make the spreadsheet look organised. Sorting can make it easy to analyse and investigate data.

For example:

We have a list of students' names. The list needs to arranged alphabetically to assign roll numbers. We use Sort to arrange the names.

Student Name	Roll no.
Nisha	
Abhishek	
Jayant	
Zoya	
Ritesh	
Dipti	

Before Sorting

Student Name	Roll no.
Abhishek	1
Dipti	2
Jayant	3
Nisha	4
Ritesh	5
Zoya	6

After Sorting

Filter

- A Filter is used to narrow down data in a spreadsheet. This lets us view only the data we need.
- Filter can be used to quickly find certain values in a spreadsheet.
- At times, we have a large set of data. Filter can be used to hide unwanted data and only display data which we require.
- Filter does not delete or modify any of the data. It only affects which data is displayed.

For example:

We have a list of pet animals which students have at home. We will apply a filter to view the most common pet animal.

Name	Pet
Abhishek	Dog
Dipti	Cat
Jayant	Dog
Neha	Dog
Nisha	Dog
Ritesh	Cat
Zoya	Parrot

Without Filter

Name	Pet
Abhishek	Dog
Jayant	Dog
Neha	Dog
Nisha	Dog

With Filter

Exercise

Fill in the blanks

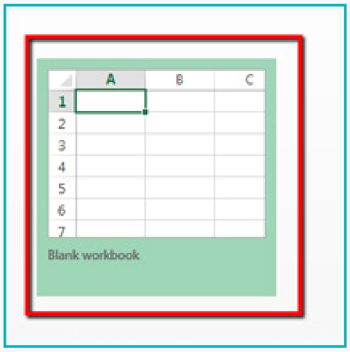
1. ______________________ lets us apply specific formatting to cells if a certain condition is true.
2. ________________ is a process that arranges data into a meaningful order.
3. ________________ are used to narrow down data in a spreadsheet and display only the data we need.

Activity

Rohan wants to plan his own birthday party. His mother has given him a budget of Rs. 2000. He has spoken to 3 different vendors to get estimates for party supplies. Let's help Rohan pick a supplier whose estimate matches his budget.

Cost of party supplies:		
Mahesh:	**Rajesh:**	**Sagar:**
Pizza Slices: Rs. 1200 Cake: Rs. 350 Balloons: Rs. 120 Ribbons: Rs. 100 Chocolate shakes: Rs. 550	Pizza Slices: Rs. 1000 Cake: Rs. 300 Balloons: Rs. 130 Ribbons: Rs. 110 Chocolate shakes: Rs. 600	Pizza Slices: Rs. 900 Cake: Rs. 400 Balloons: Rs. 100 Ribbons: Rs. 120 Chocolate shakes: Rs. 500

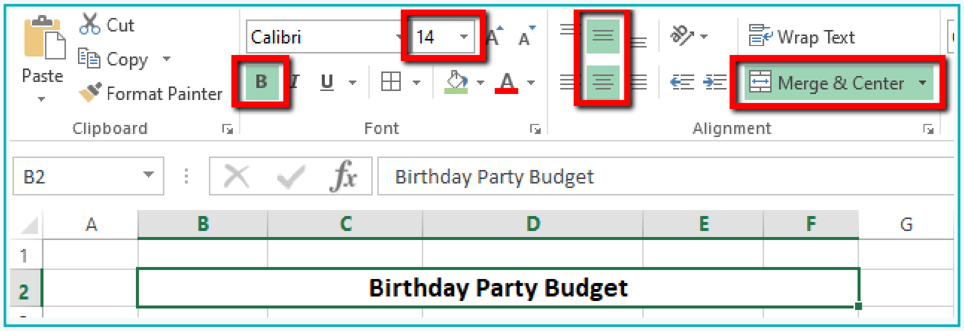
1. Launch the Excel software and click on **Blank workbook** to create a new spreadsheet.



2. Create a table to add the budget for the party:

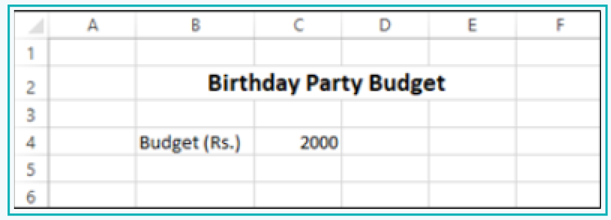
a. Add a title for the table – **Birthday Party Budget** – in Bold with **14** as the font size.

b. Select cells from **B10 to E10** and click on **Merge & Centre.**



Activity

3 Below the title, enter the budget for the birthday party. Type **Budget (Rs.)** in cell **B4.** Add the budget value **Rs. 2000** in **C4.** Resize the columns to fit the data – in column B, the text is longer than the column width.

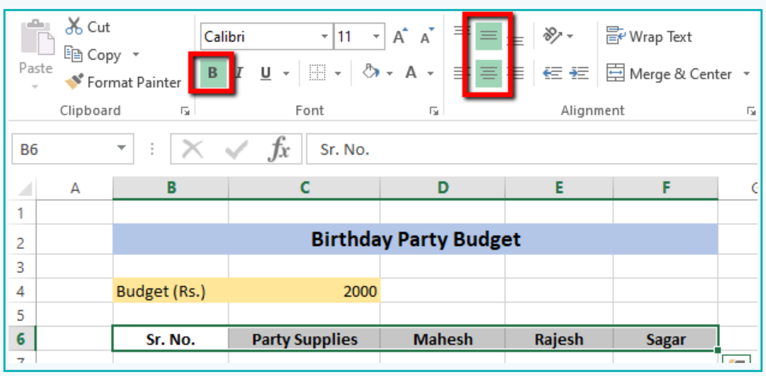


	A	B	C	D	E	F
1						
2		Birthday Party Budget				
3						
4		Budget (Rs.)	2000			
5						
6						

4 Columns are used to write categories. In this case, the categories are: Sr. No, Party Supplies, Mahesh, Rajesh, and Sagar.

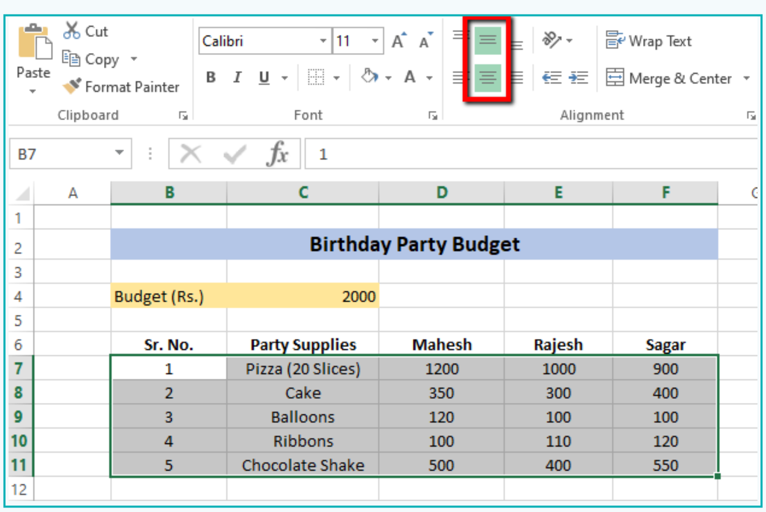
a. Format the categories to bold and centre aligned as these are headings for the entries.

b. Add background color to the headings.



	A	B	C	D	E	F
1						
2		Birthday Party Budget				
3						
4		Budget (Rs.)	2000			
5						
6		Sr. No.	Party Supplies	Mahesh	Rajesh	Sagar

5 Rows are used to add entries. In this case, the entries are: types of party supplies and their prices received from 3 different vendors.

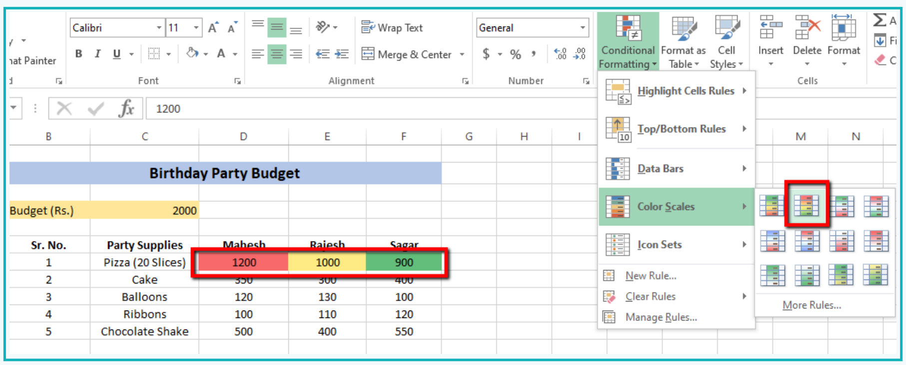


	A	B	C	D	E	F
1						
2		Birthday Party Budget				
3						
4		Budget (Rs.)	2000			
5						
6		Sr. No.	Party Supplies	Mahesh	Rajesh	Sagar
7		1	Pizza (20 Slices)	1200	1000	900
8		2	Cake	350	300	400
9		3	Balloons	120	100	100
10		4	Ribbons	100	110	120
11		5	Chocolate Shake	500	400	550
12						

Activity

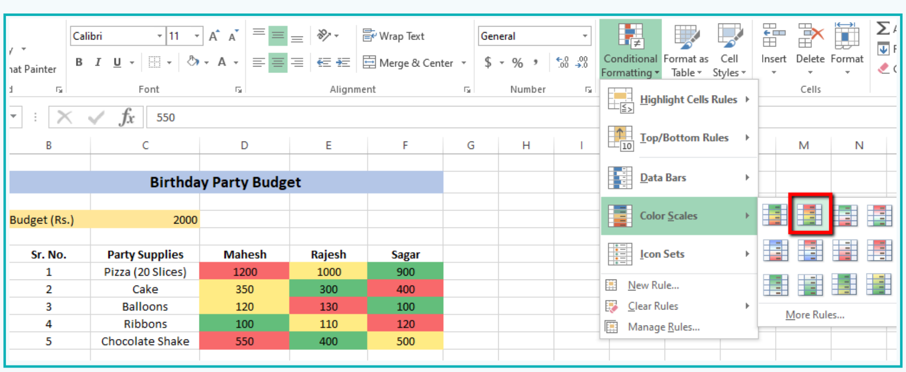
We need to help Rohan to find out whose costs for the party supplies are the least. This will help him decide who to choose according to his budget of Rs 2000. We will use conditional formatting to find this out.

6 Select all three prices of Pizza (20 pieces) and choose **Color Scales >> Red–Yellow–Green Color Scale.**



- The **red** cell shows the item that is **most costly.**
- The **yellow** cell shows the item that is **moderately costly.**
- The **green** cell shows the item that is **least costly.**

7 Apply **Red–Yellow–Green Color Scale** to all other party supplies one-by-one.

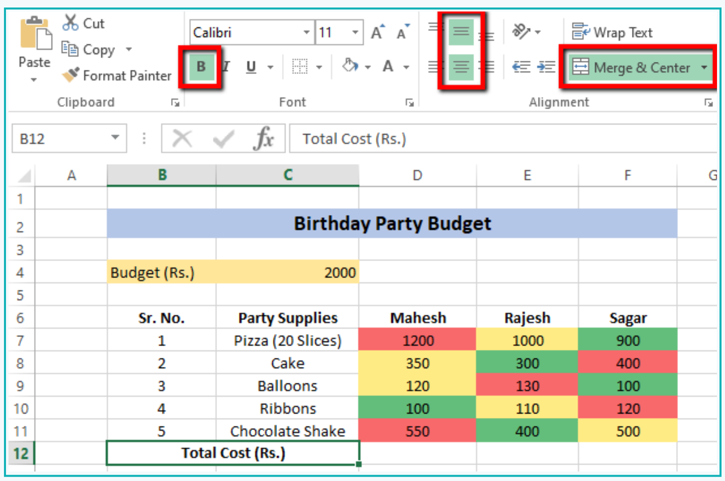


Activity

8. To identify whose cost for all the supplies matches Rohan's budget, we will calculate the total costs from Mahesh, Rajesh and Sagar using **Sum()** function.

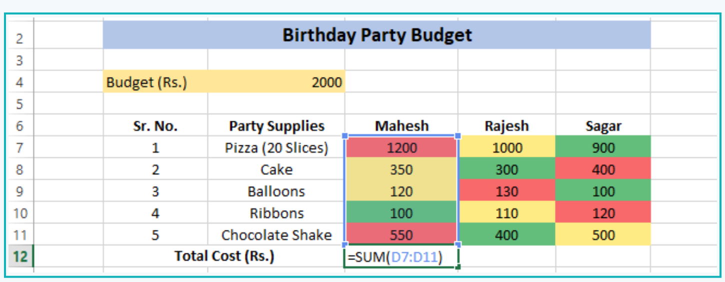
a. Enter **Total Cost (Rs.)** in cell **B12** and make it bold and centre aligned.

b. **Merge & centre** the cells **B12** and **C12** since there are no further entries in those categories.



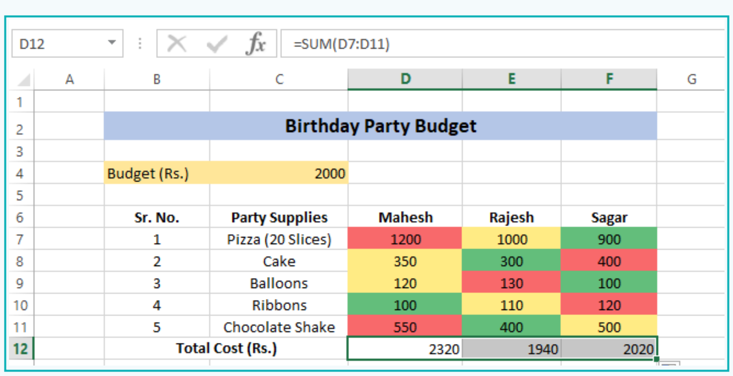
	A	B	C	D	E	F
2		Birthday Party Budget				
4		Budget (Rs.)	2000			
6		Sr. No.	Party Supplies	Mahesh	Rajesh	Sagar
7		1	Pizza (20 Slices)	1200	1000	900
8		2	Cake	350	300	400
9		3	Balloons	120	130	100
10		4	Ribbons	100	110	120
11		5	Chocolate Shake	550	400	500
12		Total Cost (Rs.)				

c. To calculate the cost of all items under Mahesh, apply the function **=SUM(D7:D11)** in cell **D12**.



	B	C	D	E	F
2	Birthday Party Budget				
4	Budget (Rs.)	2000			
6	Sr. No.	Party Supplies	Mahesh	Rajesh	Sagar
7	1	Pizza (20 Slices)	1200	1000	900
8	2	Cake	350	300	400
9	3	Balloons	120	130	100
10	4	Ribbons	100	110	120
11	5	Chocolate Shake	550	400	500
12	Total Cost (Rs.)		=SUM(D7:D11)		

d. Drag the bottom right corner of cell D12 to cells E12 and F12. This will apply the same formula to those cells.

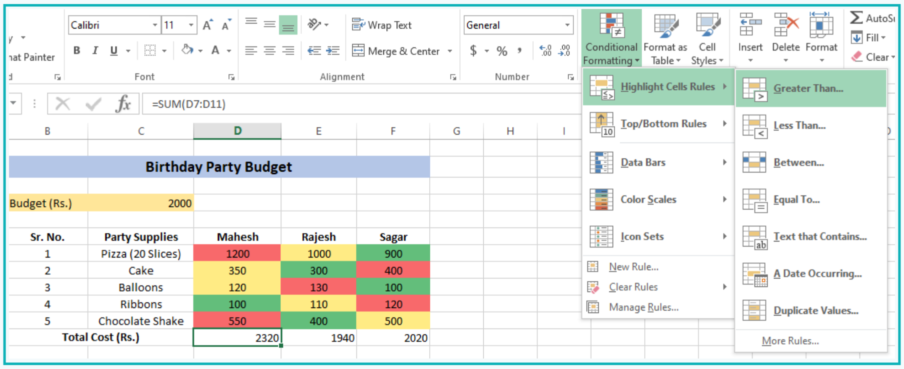


	A	B	C	D	E	F	G
2		Birthday Party Budget					
4		Budget (Rs.)	2000				
6		Sr. No.	Party Supplies	Mahesh	Rajesh	Sagar	
7		1	Pizza (20 Slices)	1200	1000	900	
8		2	Cake	350	300	400	
9		3	Balloons	120	130	100	
10		4	Ribbons	100	110	120	
11		5	Chocolate Shake	550	400	500	
12		Total Cost (Rs.)		2320	1940	2020	

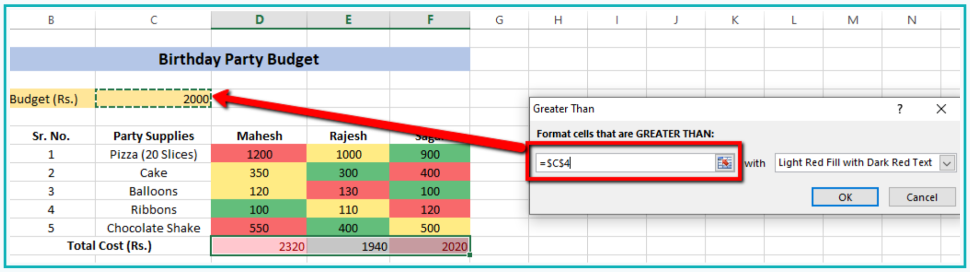
Activity

9 Apply conditional formatting to check if the total cost is above budget:

a. Select cells **D12 to F12** and click on
Conditional Formatting >> Highlight Cell Rules >> Greater Than.

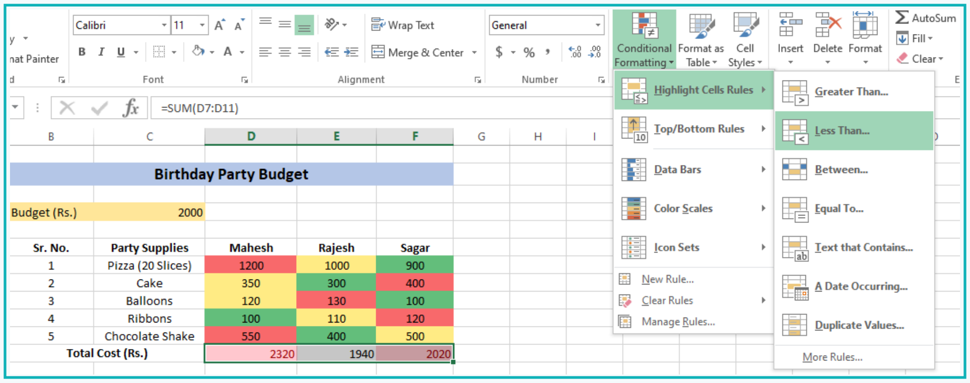


b. If the total cost of any of the cells D12, E12 or F12 is greater than the budget in cell C4, then change the cell colours to **red**.



10 Apply conditional formatting to check if the total cost is within budget:

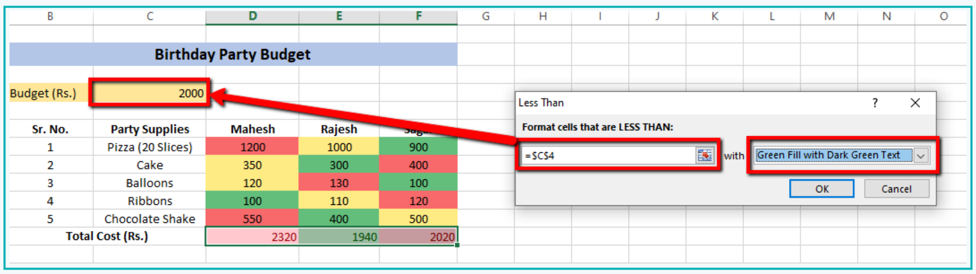
a. Select cells **D12 to F12** and click on
Conditional Formatting >> Highlight Cell Rules >> Less Than.



ICT 360

Activity

b. If the total cost of any of the cells D12, E12 or F12 is less than the budget in cell C4, then change cell colour to green:



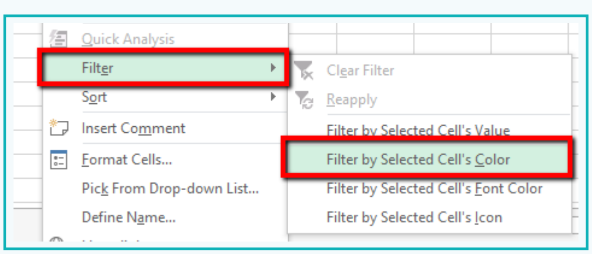
11 To separate each entry, apply a border to the table.

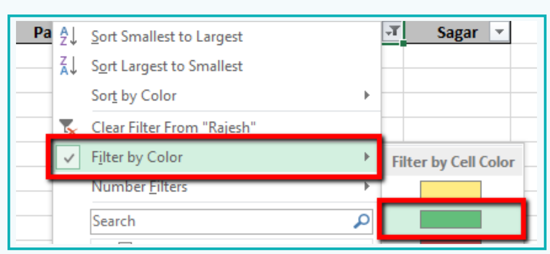
Birthday Party Budget				
Budget (Rs.)	2000			
Sr. No.	**Party Supplies**	**Mahesh**	**Rajesh**	**Sagar**
1	Pizza (20 Slices)	1200	1000	900
2	Cake	350	300	400
3	Balloons	120	130	100
4	Ribbons	100	110	120
5	Chocolate Shake	550	400	500
Total Cost (Rs.)		2320	1940	2020

We applied conditional formatting on the costs of party supplies from each store. Through this, we found that Rajesh's prices match Rohan's budget of Rs 2000.

Now let's find out which party supplies cost the lowest by applying Filter.

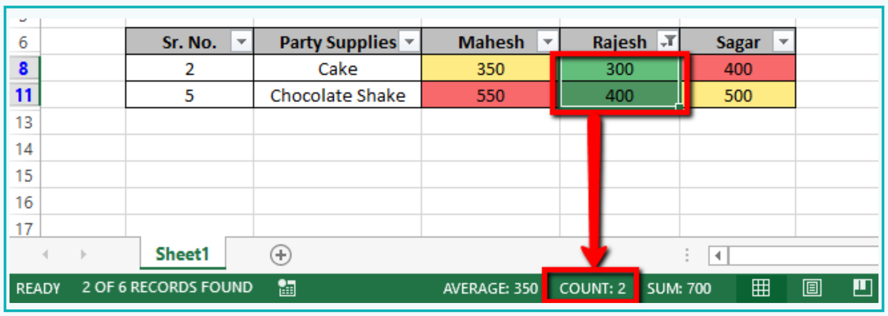
12 To identify which supplies cost is offered lowest by Rajesh, apply colour filter as shown.



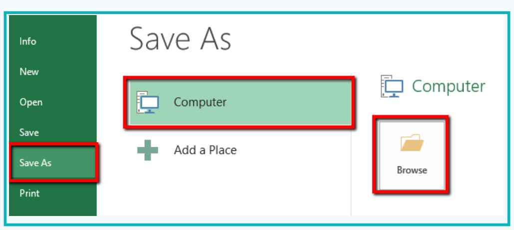


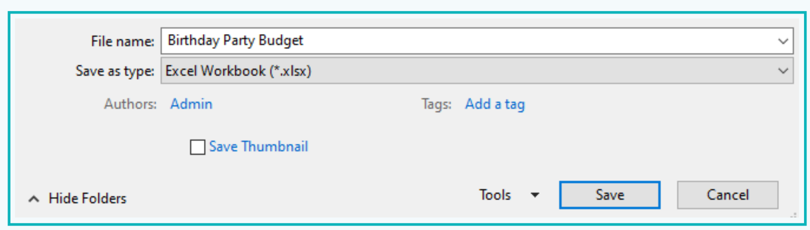
Activity

13 To view how many of Rajesh's supplies cost the least, select the cells with the green background and check the 'count' value at the bottom.

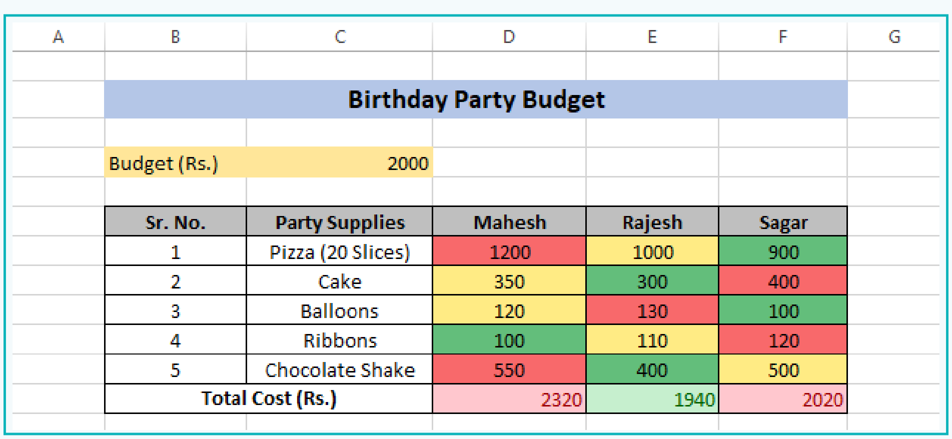


14 Save the file with the name **Birthday Party Budget.**





Final Output



Birthday Party Budget

Budget (Rs.) 2000

Sr. No.	Party Supplies	Mahesh	Rajesh	Sagar
1	Pizza (20 Slices)	1200	1000	900
2	Cake	350	300	400
3	Balloons	120	130	100
4	Ribbons	100	110	120
5	Chocolate Shake	550	400	500
Total Cost (Rs.)		2320	1940	2020

We applied conditional formatting to the costs of party supplies. We found that Rajesh's prices match Rohan's budget of Rs 2000. By applying a filter, we found that Rajesh's store provided the maximum number of party supplies at a low cost. Rohan can spend money on supplies from Rajesh for his birthday party.

Exercise

State whether True or False

4 Sort is used to automatically apply formatting to cells when the data meets a specific criteria.

▲ ______________________

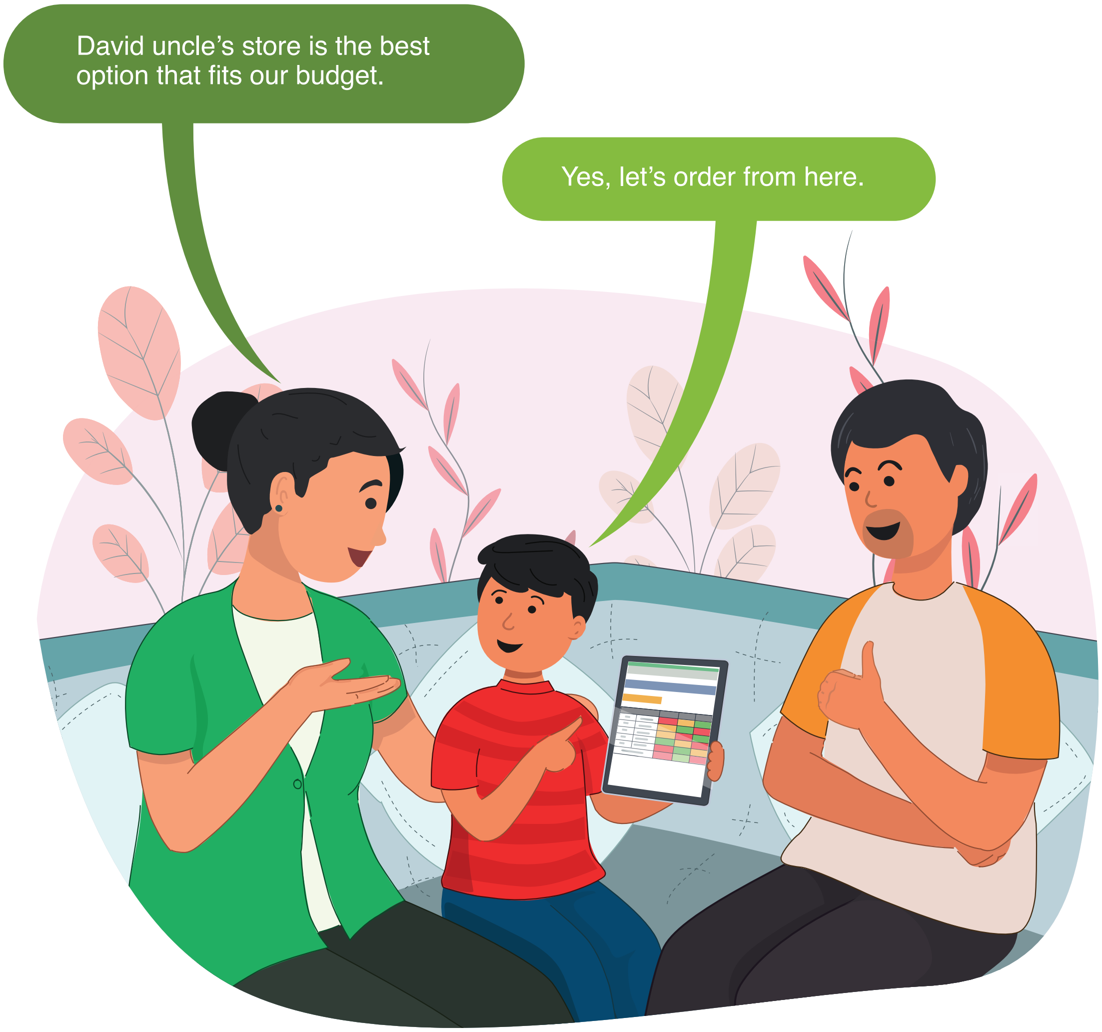
5 Filter does not remove or modify data. It just changes which data is displayed on the screen.

▲ ______________________

Spreadsheets are used to arrange large sets of data in a way that is easier to understand.

- Conditional formatting automatically applies formatting to a cell when the data meets a specific criteria.
- Sort helps us to quickly organise data alphabetically or numerically.
- Filter helps to view only relevant data.

15. Analyse Data



Tech Flash

- **Conditional formatting** : It applies specific formatting to a cell if a certain condition is true. This makes spreadsheets easier to understand.
- **Sort** : It is a process through which data is arranged into some meaningful order to make it easier to understand, analyse or visualise.
- **Filter** : It is used to narrow down data in our worksheet and display only the data that we need.